

Module 8

Technical Workouts

1. Understand Data Preparation Fundamentals

- a. Why data preparation matters
- b. Raw vs processed data
- c. Data quality challenges
- d. ML-ready datasets
- e. Data preparation lifecycle

2. Understand Data Cleaning

- a. Missing values
- b. Duplicate records
- c. Invalid entries
- d. Outlier identification
- e. Data consistency improvements

3. Understand Data Transformation

- a. Feature transformation
- b. Scaling
- c. Normalization
- d. Encoding categorical variables
- e. Data standardization

4. Understand Feature Preparation

- a. Feature creation
- b. Feature selection awareness
- c. Removing irrelevant information
- d. Improving model readiness
- e. Data quality improvements

5. Understand Data Integration

- a. Combining datasets
- b. Data merging
- c. Data aggregation
- d. Working with multiple sources
- e. Dataset consistency

6. Understand Data Pipeline Thinking

- a. Data ingestion
- b. Data cleaning workflows
- c. Transformation workflows
- d. Reusable processing pipelines
- e. Automation awareness

7. Understand Data Leakage Awareness

- a. What is data leakage
- b. Common leakage scenarios
- c. Avoiding invalid datasets
- d. Reliable ML workflows
- e. Validation awareness